Notes: Chevalier and Scharfstein (AER, 1996)

Capital-Market Imperfections and Countercyclical Markups: Theory and Evidence

Contents

1 Big picture

One-sentence summary. This note summarizes Chevalier and Scharfstein (1996), which develops a product-market model in which capital-market imperfections make financially constrained firms raise prices during downturns, and then tests this mechanism using supermarket price data around the 1986 oil shock and the 1990–1991 recession.

The paper asks why output prices can rise relative to wages and raw-material prices during recessions. In many standard aggregate-demand models, booms should make factor prices fall relative to output prices because marginal products decline at high output. The empirical fact is often the opposite: real factor prices are procyclical. One way to reconcile this fact with demand-driven cycles is that markups of price over marginal cost are countercyclical: firms behave more competitively in booms and less competitively in recessions.

Chevalier and Scharfstein focus on one mechanism: **capital-market imperfections**. The key idea is simple. In customer-market or switching-cost models, firms may keep current prices low to build market share and harvest future profits from repeat customers. But if firms are financially constrained, especially in recessions, they may care more about current cash flow and less about future market share. They therefore raise prices, increase current margins, and cut back on implicit investment in customer relationships.

The empirical setting is the U.S. supermarket industry. This setting is useful because supermarkets compete in many local markets, allowing the authors to study a single industry while exploiting cross-city variation in local demand shocks and cross-firm variation in financial constraints. The paper uses three empirical episodes: the oil-price collapse of 1986, the 1990–1991 macro recession, and post-recession firm-level scanner-price data.

2 Incremental contribution of the paper

- **Mechanism contribution.** The paper provides a clear theoretical mechanism linking financial constraints to countercyclical markups. It connects customer-market pricing with imperfect capital markets and shows that constrained firms price less aggressively in recessions.
- **Identification contribution.** Rather than estimating markups directly, which requires difficult marginal-cost assumptions, the authors compare price changes across firms and markets whose liquidity exposure should differ after plausibly exogenous shocks.
- **Empirical contribution.** The paper gives industry-level and firm-level evidence from supermarkets. Prices fall less, or rise more, when financially constrained firms have greater market presence, especially in local markets hit harder by downturns.
- **Macro-finance contribution.** The paper links financial frictions not only to investment, inventory, or employment decisions, but also to pricing behavior. This gives a channel through which financial constraints can amplify demand shocks.
- **Strategic-interaction contribution.** Because prices are strategic complements, even unconstrained firms raise prices when their rivals are constrained. Financial shocks can therefore spread through product-market competition.

Increment relative to the existing countercyclical-markups literature. Rotemberg and

Saloner-style models emphasize weaker collusion in booms. Chevalier and Scharfstein emphasize a different force: liquidity-constrained firms increase current margins because they cannot afford to invest in market share. Their evidence is difficult to reconcile with the pure tacit-collusion explanation because liquidity-constrained firms should have more incentive to cut prices in recessions under a collusion model, whereas the data show that they raise prices relative to unconstrained firms.

3 Research question and conceptual framework

The central research question is:

Do capital-market imperfections make product-market markups more countercyclical?

More specifically, the paper asks whether financially constrained firms in downturns raise prices relative to financially unconstrained firms. The conceptual distinction is between:

- **Current cash-flow motive:** a constrained firm needs cash now and therefore increases prices to raise short-run profits.
- **Market-share investment motive:** an unconstrained firm is willing to keep prices low today to attract customers and earn higher future profits from repeat business.

The paper studies markups indirectly through price changes. This avoids the difficult task of estimating marginal costs. The empirical idea is that if two kinds of supermarket chains face similar marginal-cost movements but different liquidity shocks, then relative price increases among more constrained chains can be interpreted as relative markup increases.

4 The model: customer markets with financial constraints

4.1 Product-market environment

The product-market model builds on a two-period switching-cost framework. Two firms, A and B, compete for customers. Consumers are distributed on a line segment $[0, 1]$, with firm A located at 0 and firm B located at 1. Consumers choose between firms in period 1 and become partly locked in because switching in period 2 is costly.

Let the first-period price of firm A be p^A and that of firm B be p^B. A consumer located at y buys from firm A rather than firm B if:

$$p^A + ty \leq p^B + t(1 - y),$$

which implies firm A's first-period market share:

$$\sigma_1^A = \frac{1}{2} + \frac{p^B - p^A}{2t}, \qquad \sigma_1^B = 1 - \sigma_1^A.$$

In period 2, firms can charge locked-in customers high prices because switching is costly. If the switching cost is sufficiently high, firm k's period-2 profit is:

$$\pi_2^k(\sigma_1^k) = (R - c_2)\sigma_1^k.$$

Thus, first-period market share is an investment asset. A low current price is costly today but valuable tomorrow.

4.2 Benchmark with internal finance

When firms are internally financed, firm A maximizes the present value of first- and second-period profits:

$$(p^A - c_1)\theta_i \sigma_1^A + (R - c_2)\sigma_1^A.$$

The reaction curve is:

$$p^A = \frac{t + c_1}{2} + \frac{p^B}{2} - \frac{R - c_2}{2\bar{\theta}}.$$

The symmetric equilibrium price when both firms are internally financed is:

$$p^*(0,0) = t + c_1 - \frac{R - c_2}{\bar{\theta}}.$$

The term $-(R - c_2)/\bar{\theta}$ captures market-share investment. Firms price below the static one-period price because lower prices today increase future locked-in profits.

In the benchmark version, markups are procyclical. A boom raises current demand relative to future demand, so it becomes less attractive to sacrifice current profits for future market share.

4.3 Introducing capital-market imperfections

The authors then introduce an imperfect external finance problem. Firms must raise an upfront investment I. Cash flow is observable to managers but not verifiable to outside investors. The manager can divert cash flow. Investors can discipline the manager only by threatening liquidation, but liquidation is inefficient.

The optimal contract resembles debt. The firm must pay D at date 1. If the firm cannot pay, investors can seize and liquidate the project. This changes pricing incentives. If a firm may be liquidated in bad states, it cares less about building market share because it may not survive to enjoy future customer rents.

With external finance, firm A's reaction curve becomes:

$$p^A = \frac{t + c_1}{2} + \frac{p^B}{2} - \mu \frac{R - c_2}{2\bar{\theta}},$$

where μ is the probability of the high-demand state. Since $\mu < 1$, the market-share-investment term is smaller in magnitude. Therefore, externally financed firms charge higher prices than internally financed firms.

If both firms are externally financed, the symmetric equilibrium price is:

$$p^*(1,1) = t + c_1 - \frac{\mu}{\bar{\theta}}(R - c_2),$$

which is higher than $p^*(0,0)$ when $\mu < 1$.

4.4 Strategic complements and rival financial constraints

Prices are strategic complements. When one firm becomes constrained and raises its price, its rival also raises its price. This matters because the effect of financial constraints is not confined to constrained firms. It can spread to unconstrained rivals through strategic pricing.

For firm A, the model implies:

$$\frac{dm^{*A}(e_A, e_B)}{d\mu} = \left(\left[1 - \frac{2e_A + e_B}{3} \right] \theta_H - \theta_L \right) \frac{R - c_2}{\bar{\theta}}.$$

The cyclicality of markups depends on both firm A's financing status and firm B's financing status. A firm's markup becomes more countercyclical if either it or its rival is externally financed.

4.5 Three empirical predictions

The model yields three testable implications:

1. A firm's markup should be more countercyclical if the firm is more financially constrained.
2. A firm's markup should be more countercyclical if its rivals are more financially constrained.
3. Industry-wide markups should be more countercyclical if the firms in the industry are more financially constrained.

5 Empirical approach

5.1 Why the paper does not directly estimate markups

The authors do not directly estimate price-cost markups because marginal costs are hard to measure. Standard markup estimation requires strong assumptions about production functions or marginal-cost behavior. Instead, the paper studies relative price changes around shocks that differentially affect firm liquidity.

The core identifying logic can be written as:

$$\Delta p^L - \Delta p^N = (\Delta m^L - \Delta m^N) + (\Delta c^L - \Delta c^N),$$

where L denotes local or financially constrained chains and N denotes national or less constrained chains. If marginal-cost differences are unlikely to move differentially in the treated shock markets, then relative price changes reveal relative markup changes.

The paper uses three empirical designs:

- **Oil shock of 1986:** local and regional chains in oil states should be more liquidity constrained than national chains.
- **1990–1991 recession with city-level LBO exposure:** local markets dominated by LBO chains should have more financially constrained firms.
- **1991–1992 firm-level scanner prices:** LBO firms and firms facing LBO rivals should raise prices more in weak local economies.

6 Evidence 1: the 1986 oil shock

6.1 Empirical design

The oil-price collapse in early 1986 generated severe downturns in oil-producing states. The affected states include Texas, Louisiana, New Mexico, Oklahoma, Colorado, Wyoming, and Alaska. The authors use the period from 1985:4 to 1987:1, from just before the oil-price collapse to the employment trough in Texas.

The main treatment-intensity variable is **NATSHARE**, the fraction of a city's supermarket stores owned by national chains. National chains had operations outside oil states and therefore were less exposed to the local liquidity shock. Local and regional chains were more exposed.

The dependent variable is **APPRICE**, the percentage change in the local supermarket price index. Prices come from the ACCRA Cost of Living Index. Chain locations come from *Supermarket News*.

The key regressions interact **NATSHARE** with measures of local exposure to the oil shock:

- **OILDUM**: dummy for oil states.
- **OILIMP**: oil and gas extraction share of state earnings.
- **AEMP**: percentage change in state employment.

6.2 Main result

The results support the theory. Prices fall more in oil-state cities where national chains have a larger share. Equivalently, prices fall less, or rise more, where local and regional chains are more important. Since local/regional chains are more exposed to the oil shock, this is consistent with financially constrained firms raising markups in downturns.

In Table 2, the interaction **NATSHARE** $\times$ **OILDUM** is -0.115 with a t-statistic of -2.270. The authors interpret this as evidence that in oil states, a higher national-chain share is associated with larger price declines. The coefficient is economically meaningful: moving NATSHARE one standard deviation above its mean in an oil-state city reduces the expected price change substantially.

The results are similar when using continuous oil exposure and employment declines. In cities more exposed to oil or with larger employment losses, national-chain dominance predicts greater price declines; local-chain dominance predicts smaller price declines.

7 Evidence 2: LBO supermarket chains and the 1990–1991 recession

7.1 Empirical design

The second test uses the 1990–1991 recession. During the 1980s, many supermarket chains undertook leveraged buyouts. These LBOs increased leverage and made the chains more cash-flow constrained. The authors therefore predict that prices should fall less or rise more in cities where LBO chains have greater market share, especially in cities hit harder by the recession.

The key variable is **LBOSHARE**, the fraction of stores in a city owned by chains that had undertaken LBOs before 1990:2. The dependent variable is the percentage change in the local price index from 1990:2 to 1991:1.

7.2 Main result

The coefficient on **LBOSHARE** alone is not statistically significant, but the interaction **LBOSHARE** $\times$ **AEMP** is negative and significant at roughly the 8 percent level. Since AEMP is lower in worse local recessions, the negative interaction means that LBO-chain exposure raises prices more in cities with weaker employment growth.

The economic magnitude is meaningful. In a city with employment growth one standard deviation below the mean, increasing LBOSHARE by one standard deviation more than doubles the expected price increase from about 0.8 percent to 1.8 percent.

8 Evidence 3: firm-level scanner prices after the 1990–1991 recession

8.1 Empirical design

The third test uses firm-level pricing data from Information Resources, Inc. (IRI). These scanner data allow the authors to compare the prices of individual supermarket firms in local markets. This is the most direct test of the model's firm-level predictions.

The dependent variable is the percentage change in a firm's local-market price index. The key explanatory variables are:

- **LBO**: dummy equal to one if the firm had undergone an LBO.
- **LBO** $\times$ **AEMP**: whether LBO firms raise prices more in weaker local economies.
- **OLBOSHARE**: the share of local rival stores owned by LBO chains.
- **OLBOSHARE** $\times$ **AEMP**: whether rival LBO exposure matters more in weaker local economies.

The authors study two horizons: 1991:1–1991:4 and 1991:1–1992:4.

8.2 Main result

The results support both firm-level implications of the model. First, LBO firms raise prices more than non-LBO firms, especially in weak local economies. Second, firms raise prices more when their rivals are LBO firms, especially in weak local economies.

In Table 6, the coefficient on **LBO** is positive and statistically significant in both horizons. The coefficient on **LBO** $\times$ **AEMP** is negative in both horizons and statistically stronger for the shorter 1991:1–1991:4 period. Because AEMP is lower in weak markets, this means LBO firms raise prices more relative to non-LBO firms where local conditions are weaker.

The coefficient on **OLBOSHARE** is also positive and significant in both horizons. This shows that firms raise prices more when rivals are financially constrained. The short-horizon coefficient on **OLBOSHARE** $\times$ **AEMP** is negative and significant, consistent with strategic complementarities in prices.

9 Identification and interpretation

9.1 Why the evidence supports the financial-constraints mechanism

The empirical evidence is persuasive because it uses multiple shocks and multiple measures of financial constraints. The oil shock identifies local liquidity exposure through regional chain concentration. The 1990–1991 recession identifies financial constraints through LBO status. The IRI scanner data identify firm-level pricing differences and rival effects.

The key identification idea is not that all price changes are markup changes. Instead, the argument is that the differential price response across constrained and unconstrained chains is hard to explain through differential marginal-cost movements alone, especially because the interaction effects are strongest in markets hit hardest by demand shocks.

9.2 Why the results challenge the tacit-collusion interpretation

The results do not fit well with a pure Rotemberg-Saloner collusion story. In that model, recessions make firms less tempted to cheat on collusive arrangements, so markups can rise in recessions. But if a firm is liquidity constrained, it should have a stronger short-run incentive to cut prices and steal market share, not to raise prices. Chevalier and Scharfstein instead find that constrained firms raise prices relative to unconstrained firms. This supports the market-share-investment mechanism rather than the countercyclical-collusion mechanism.

9.3 Why strategic complements matter

The firm-level evidence on OLBOSHARE is important because it shows that financial constraints can affect not only constrained firms but also their rivals. If a constrained firm raises price, competitors can optimally raise price too. Thus, a liquidity shock to one group of firms can transmit through the product market and affect industry-level pricing.

10 Extracted figures and tables from the paper

10.1 Figure 1. Reaction Curves for Firms A and B with Internal and External Financing

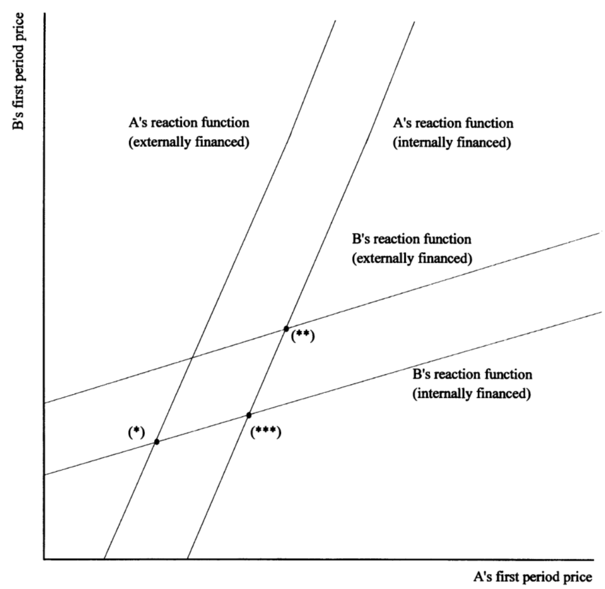

FIGURE 1. REACTION CURVES FOR FIRMS A AND B WITH INTERNAL AND EXTERNAL FINANCING

Figure 1. Reaction Curves for Firms A and B with Internal and External Financing

Read. Figure 1 illustrates how external financing shifts the reaction curves of firms. When firm A is externally financed, its reaction curve shifts so that for a given rival price, it charges a higher price. When firm B is externally financed, its reaction curve shifts similarly. The equilibrium with both firms externally financed is at a higher price than the equilibrium with internal finance.

Economic message. Financial constraints reduce the incentive to cut prices to build market share. Because prices are strategic complements, one firm's financial constraint also raises its rival's optimal price.

10.2 Table 1. Summary Statistics: The Oil Shock of 1986

10.3 Table 2. Regression Results: The Oil Shock of 1986

TABLE 1—SUMMARY STATISTICS: THE OIL SHOCK OF 1986

Variable	Mean	Variable	Mean
ΔPRICE	−0.006 (0.049)	NATSHARE × OILIMP	0.005 (0.008)
NATSHARE	0.347 (0.229)	ΔEMP	−0.003 (0.033)
OILDUM	0.220	NATSHARE × ΔEMP	−0.002 (0.014)
NATSHARE × OILDUM	0.109 (0.232)	ΔWAGE	0.058 (0.389)
OILIMP	0.011 (0.017)		

Notes: This table reports summary statistics for the sample of 100 observations analyzed in Table 2. ΔPRICE is the percentage change in the local supermarket price index over the period 1985:4 to 1987:1. NATSHARE is the fraction of a city's stores that are owned by national chains. OILIMP is the share of a state's earnings accounted for by oil and gas production. OILDUM is a dummy variable that takes the value 1 if the city is in a state in which OILIMP is greater than 2 percent. ΔEMP is the percentage change in employment in the city's state over this period. ΔWAGE is the percentage change in wages of a sample of workers in sales occupations according to the *Current Population Survey*. Standard deviations are reported in parentheses below the means.

TABLE 2—REGRESSION RESULTS: THE OIL SHOCK OF 1986

	Coefficients		
Variable	(1)	(2)	(3)
NATSHARE	0.007 (0.305)	−0.006 (−0.223)	−0.038 (−1.852)
NATSHARE × OILDUM	−0.115 (−2.270)		
OILDUM	0.017 (0.651)		
NATSHARE × OILIMP		−3.033 (1.724)	
OILIMP		0.721 (0.964)	
NATSHARE × ΔEMP			2.267 (2.365)
ΔEMP			−0.641 (−1.570)
ΔWAGE	0.007 (0.59)	0.008 (0.641)	0.007 (0.565)
Constant	0.000 (0.000)	0.002 (0.200)	0.009 (0.999)
$\bar{R}^2$	0.170	0.102	0.130

Notes: This table reports regression results in which the dependent variable is ΔPRICE, the percentage change in a city's price index from 1985:4 to 1987:1. NATSHARE is the fraction of a city's stores that are owned by national chains. OILIMP is the share of a state's earnings accounted for by oil and gas production. OILDUM is a dummy variable that takes the value 1 if the city is in a state in which OILIMP is greater than 2 percent. ΔEMP is the percentage change in employment in the city's state over the period. ΔWAGE is the percentage change in wages of a sample of workers in sales occupations according to the *Current Population Survey*. There are 100 observations. *t* statistics are in parentheses

Table 1. Summary Statistics: The Oil Shock of 1986

Table 2. Regression Results: The Oil Shock of 1986

Read. Table 1 reports the variables used in the oil-shock analysis, including the city-level price change, national-chain store share, oil-state indicators, oil exposure, state employment growth, and wage changes. Table 2 reports regressions of the local price change on national-chain share and interactions between national-chain share and local shock measures.

Economic message. The important coefficients are the interactions involving NATSHARE. The negative coefficient on NATSHARE × OILDUM and the positive coefficient on NATSHARE × AEMP imply that in oil-shock markets, prices fall more when national chains have higher market share. Therefore prices fall less where local and regional, more liquidity-constrained chains dominate.

10.4 Table 3. Summary Statistics: 1990–1991 Recession

10.5 Table 4. Regression Results: 1990–1991 Recession

TABLE 3—SUMMARY STATISTICS: 1990–1991 RECESSION

Variable	Mean
ΔPRICE	0.006
	(0.029)
LBOSHARE	0.246
	(0.194)
ΔEMP	−0.022
	(0.016)
LBOSHARE × ΔEMP	−0.005
	(0.006)
ΔWAGE	0.035
	(0.236)

Notes: This table reports summary statistics for the sample of 59 observations analyzed in Table 4. ΔPRICE, the percentage change in a city's price index from 1990:2 to 1991:1. LBOSHARE is the fraction of a city's stores that are owned by chains that undertook a leveraged buyout during the 1980's. ΔEMP is the percentage change in employment in the city's state during the period. ΔWAGE is the percentage change in wages of a sample of workers in sales occupations according to the *Current Population Survey*. Standard deviations are reported in parentheses below the means.

Table 3. Summary Statistics: 1990–1991 Recession

TABLE 4—REGRESSION RESULTS: 1990–1991 RECESSION

Variable	Coefficient
LBOSHARE	−0.024
	(0.788)
LBOSHARE × ΔEMP	−1.970
	(1.767)
ΔEMP	0.384
	(1.005)
ΔWAGE	0.0173
	(1.052)
Constant	0.010
	(0.864)
$\bar{R}^2$	0.095

Notes: This table reports regression results where the dependent variable is ΔPRICE, the percentage change in a city's price index from 1990:2 to 1991:1. LBOSHARE is the fraction of a city's stores that are owned by chains that undertook a leveraged buyout during the 1980's. ΔEMP is the percentage change in employment in the city's state during the period. ΔWAGE is the percentage change in wages of a sample of workers in sales occupations according to the *Current Population Survey*. There are 59

Table 4. Regression Results: 1990–1991 Recession

Read. Table 3 summarizes the variables for the 1990–1991 recession analysis: price changes, the market share of LBO chains, employment changes, and wage changes. Table 4 regresses city-level price changes on LBOSHARE, employment changes, their interaction, and wage controls.

Economic message. The interaction LBOSHARE × AEMP is negative. Since lower AEMP indicates worse local economic performance, this means LBO-chain exposure is associated with larger price increases in weaker cities. This is consistent with LBO firms being more liquidity constrained during downturns.

10.6 Table 5. Summary Statistics: 1990–1991 Recession Using Firm-Specific Data

10.7 Table 6. Regression Results: 1990–1991 Recession Using Firm-Specific Data

TABLE 5—SUMMARY STATISTICS: 1990–1991 RECESSION USING FIRM-SPECIFIC DATA

Variables	Mean 1991:1–1991:4	Mean 1991:1–1992:4
ΔPRICE	0.023	0.080
	(0.028)	(0.031)
LBO	0.382	0.393
LBO × ΔEMP	0.007	0.010
	(0.012)	(0.021)
OLBOSHARE	0.149	0.151
	(0.151)	(0.146)
OLBOSHARE × ΔEMP	0.002	0.004
	(0.004)	(0.007)
ΔEMP	0.020	0.030
	(0.015)	(0.028)
ΔWAGE	0.053	0.124
	(0.221)	(0.266)

Notes: This table reports summary statistics for the sample analyzed in Table 6. ΔPRICE is the percentage change in a firm's price index for a particular city in one of two periods, 1991:1 to 1991:4 and 1991:1 to 1992:4. LBO is a dummy variable which takes the value of one if the firm had previously undertaken a leveraged buyout. ΔEMP is the percentage change in employment in the city's state during the period. OLBOSHARE is the share of stores in the local market owned by LBO chains other than the firm itself. ΔWAGE is the percentage change in wages of a sample of workers in sales occupations according to the *Current Population Survey*. There are 110 observations in the shorter sample and 89 observations in the longer sample. Standard deviations are reported in parentheses below the means.

Table 5. Summary Statistics: 1990–1991 Recession Using Firm-Specific Data

TABLE 6—REGRESSION RESULTS: 1990–1991 RECESSION USING FIRM-SPECIFIC DATA

	Coefficients	
Variables	1991:1–1991:4	1991:1–1992:4
LBO	0.030	0.025
	(3.235)	(2.670)
LBO × ΔEMP	−1.131	−0.402
	(−2.942)	(−1.659)
OLBOSHARE	0.109	0.120
	(3.454)	(3.456)
OLBOSHARE × ΔEMP	−2.834	−1.152
	(−2.184)	(−1.389)
ΔEMP	0.712	0.375
	(2.578)	(1.807)
ΔWAGE	−0.004	−0.011
	(−0.333)	(−0.892)
Constant	−0.003	0.050
	(−0.413)	(5.044)
$\bar{R}^2$	0.179	0.200
Number of observations	110	89

Notes: This table reports regression results where the dependent variable is ΔPRICE, the percentage change in a firm's price index for a particular city over two different time periods, 1991:1, 1991:4 and 1991:1–1992:4. ΔEMP is the state employment change during the period; LBO is a dummy variable equal to 1 if the firm undertook an LBO; LBOEMP is LBO multiplied by ΔEMP; OLBOSHARE is a measure of the market shares of the other supermarket chains that undertook leveraged buyouts; OLBOEMP is OLBOSHARE multiplied by ΔEMP. There are 110 observations in the shorter sample and 89 observations in the longer sample. *t* statistics are in parentheses below the estimated coefficients.

Table 6. Regression Results: 1990–1991 Recession Using Firm-Specific Data

Read. Table 5 reports summary statistics for the IRI firm-level scanner-price sample. Table 6 estimates firm-level price changes as a function of firm LBO status, rival LBO share, local employment growth, and interactions.

Economic message. LBO firms raise prices more than non-LBO firms, and firms raise prices more when rivals are LBO firms. These effects are stronger in weaker local economies. This supports both the direct financial-constraint channel and the strategic-complement channel.

11 Relation to broader literatures

11.1 Business-cycle markups

The paper contributes to the literature on countercyclical markups by offering a financial-frictions mechanism. Unlike explanations based only on procyclical demand elasticities or collusive price wars, this mechanism operates through firms' ability and willingness to invest in future market share.

11.2 Corporate finance and product-market competition

The paper also contributes to the corporate-finance literature showing that financing structure affects real behavior. The key real outcome here is not capital expenditure, employment, or inventory, but pricing. This makes the paper an early and influential contribution to the study of how capital structure shapes product-market competition.

11.3 Financial accelerator and internal transmission

The paper's conclusion links its mechanism to financial-accelerator ideas. A negative cash-flow shock can cause constrained firms to raise markups and reduce market-share investment. Because rivals respond strategically, the shock can spread within the product market. The authors also connect this logic to evidence that conglomerates transmit shocks across divisions through internal capital markets.

12 Limitations and caveats

- **Single-industry setting.** The supermarket industry is useful for identification, but it is not clear how large the same mechanism is for the aggregate economy.
- **Indirect markup measurement.** The paper uses relative price changes rather than direct marginal-cost estimates. This is a strength for avoiding production-function assumptions, but it means the evidence supports relative markup movements indirectly.
- **Switching-cost assumption.** The model relies on firms investing in market share. The authors provide evidence that short-run supermarket demand is inelastic, but the model's mechanism may be stronger in some industries than others.
- **Financial constraint proxies.** LBO status and local-chain exposure are plausible proxies for financial constraints, but they are not direct measures of internal liquidity at the store-market level.
- **External validity.** The results come from the 1980s and early 1990s. Supermarket competition, financing conditions, and scanner-pricing strategies may differ in later periods.

13 Short summary

- The paper argues that capital-market imperfections can make markups countercyclical.
- Financially constrained firms value current cash more and future market share less.
- Therefore, in recessions, constrained firms raise prices relative to unconstrained firms.
- Strategic complementarities imply that unconstrained rivals may also raise prices when constrained rivals raise prices.
- Evidence from the supermarket industry supports the theory across three designs: the 1986 oil shock, the 1990–1991 recession, and post-recession firm-level scanner data.

14 Conclusion

Chevalier and Scharfstein (1996) show that financial constraints can affect not only investment but also pricing. In their model, firms normally cut prices to build market share and harvest future rents from repeat customers. When capital markets are imperfect and firms face liquidity pressure, especially in downturns, they reduce this market-share investment and raise prices. This generates countercyclical markups.

The empirical evidence is strongest because it is built from multiple designs. Local and regional supermarket chains in oil states raise prices relative to national chains after the 1986 oil shock. Markets with more LBO supermarket chains experience higher relative price increases during the 1990–1991 recession, especially when local employment falls more. Firm-level scanner data show that LBO firms and firms facing LBO rivals raise prices more in weak local markets.

The paper's broader takeaway is that financial frictions interact with product-market competition. Capital structure can shape pricing, markups, and the propagation of shocks. This makes the paper important for corporate finance, industrial organization, and macroeconomics.

14.1 Incremental contribution revisited

The paper's incremental contribution is to show, theoretically and empirically, that liquidity constraints can make firms choose higher current prices by reducing their investment in future customer relationships. It also shows that the financial condition of rivals matters because prices are strategic complements. This gives a channel through which financial shocks become product-market shocks.